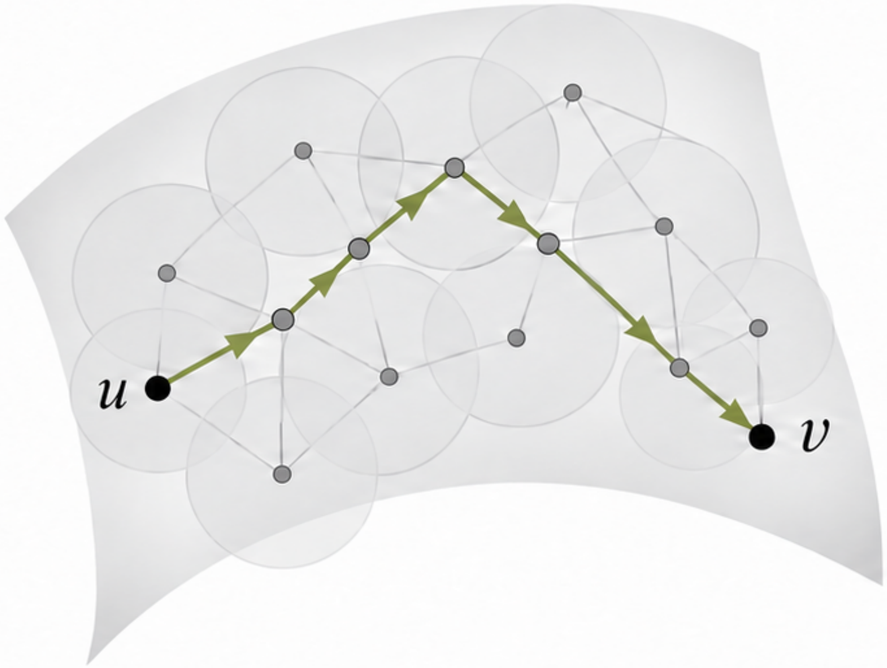
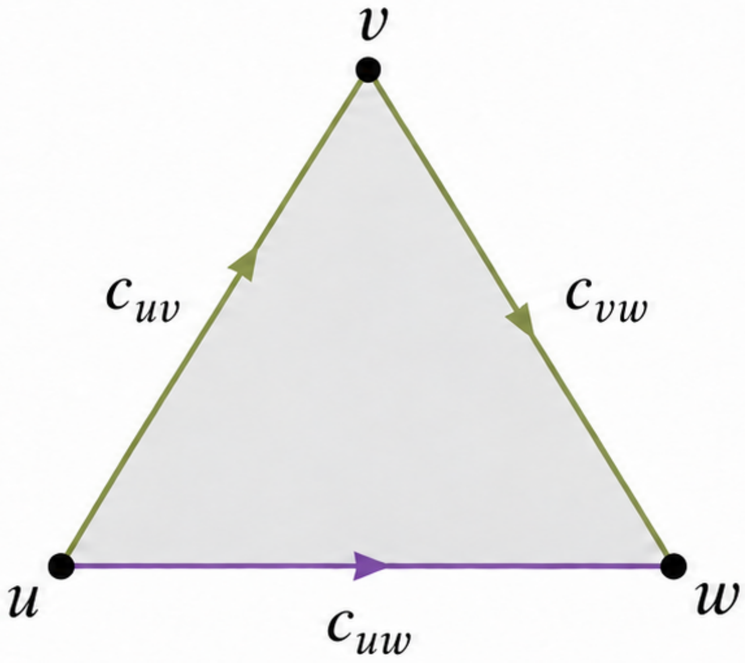
(a) Supported paths



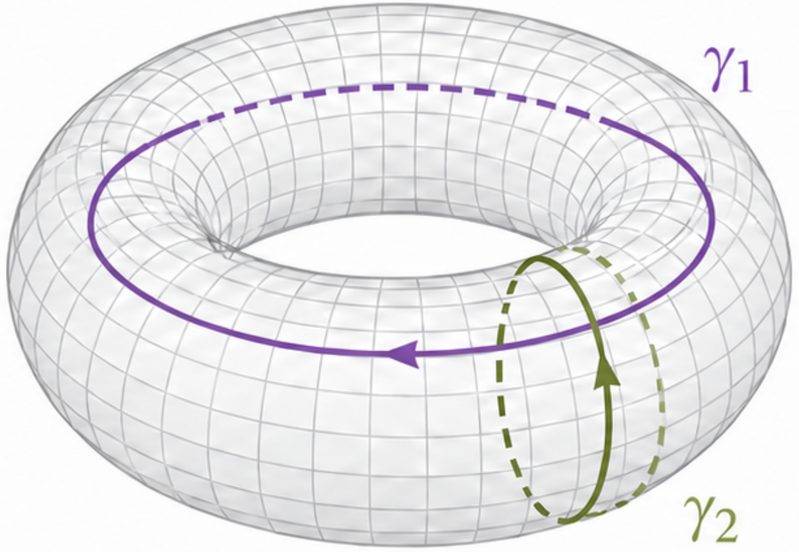
$$b_{uv} = \int_{u \to v} \beta$$

(b) Triangle closure



$$c_{uv} + c_{vw} - c_{uw} = 0$$

(c) Independent cycles



$$\sum_{(u,v)\in\gamma_j} \bar{b}_{uv} = \Pi_{:,j}$$